NeuroFlow CFD Analyst

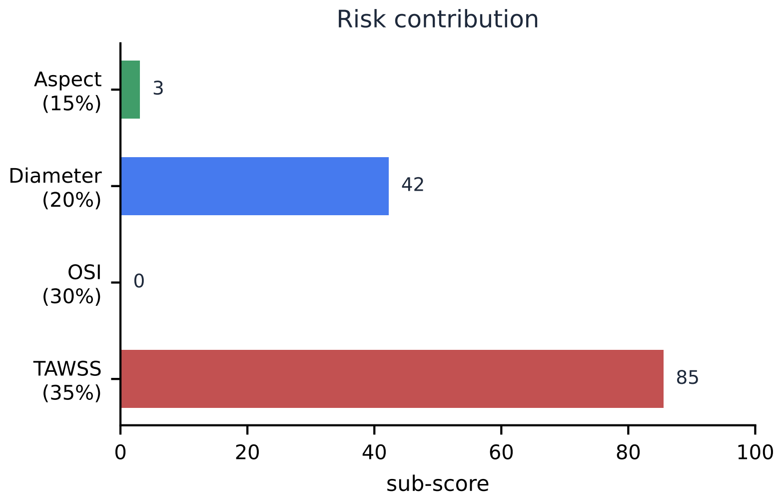
Clinical Hemodynamic Assessment

Case PT-2026-0103 · generated 2026-08-05 08:15

39

Composite Risk Index / 100

Low



Hemodynamic parameters

Parameter	Value	Threshold	Status
TAWSS — aneurysm dome	**0.346 Pa**	< 0.4 Pa	**FLAGGED**
TAWSS — parent artery	**2.939 Pa**	reference	normal
OSI — aneurysm dome	**0.008**	> 0.2	normal
RRT — aneurysm dome	**7.17 Pa^{-1}**	> 3.0	**FLAGGED**
ECAP — aneurysm dome	**0.173 Pa^{-1}**	> 1.0	normal
NWSS (sac / parent)	**0.118**	—	normal
LSAR (relative, <10% parent)	**71.3 %**	—	normal
LSAR (absolute, <0.4 Pa)	**78.3 %**	—	normal

Morphology (measured from the reconstructed surface)

Max diameter	Neck diameter	Aspect ratio	Dome-to-neck
5.38 mm	**4.83 mm**	**0.75**	**1.11**
Volume	Surface area	Non-sphericity	
63.4 mm^3	**61.1 mm^2**	**0.001**	

Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 238994 · SIMPLE solution converged in 185 iterations
WSS converted from OpenFOAM kinematic units (m^2/s^2) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution tau = 4*mu*Q/(pi*r^3).

Research software — not a medical device. Not for clinical decision-making.

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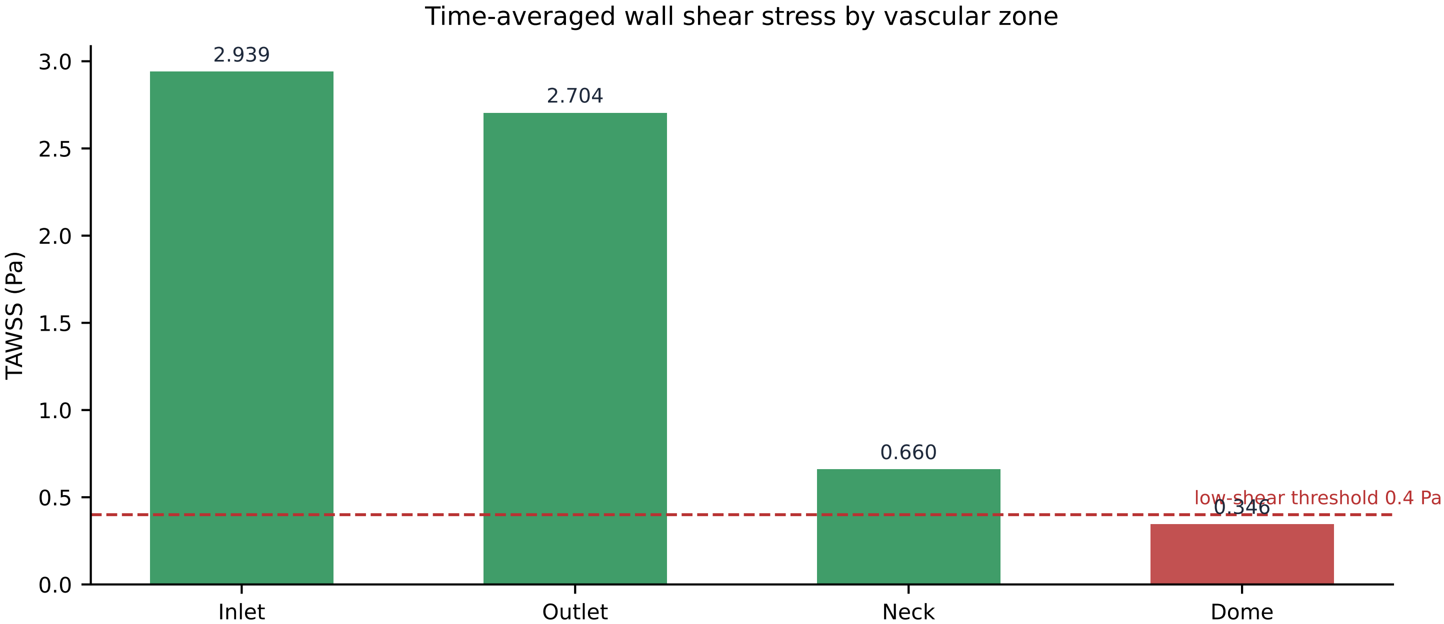
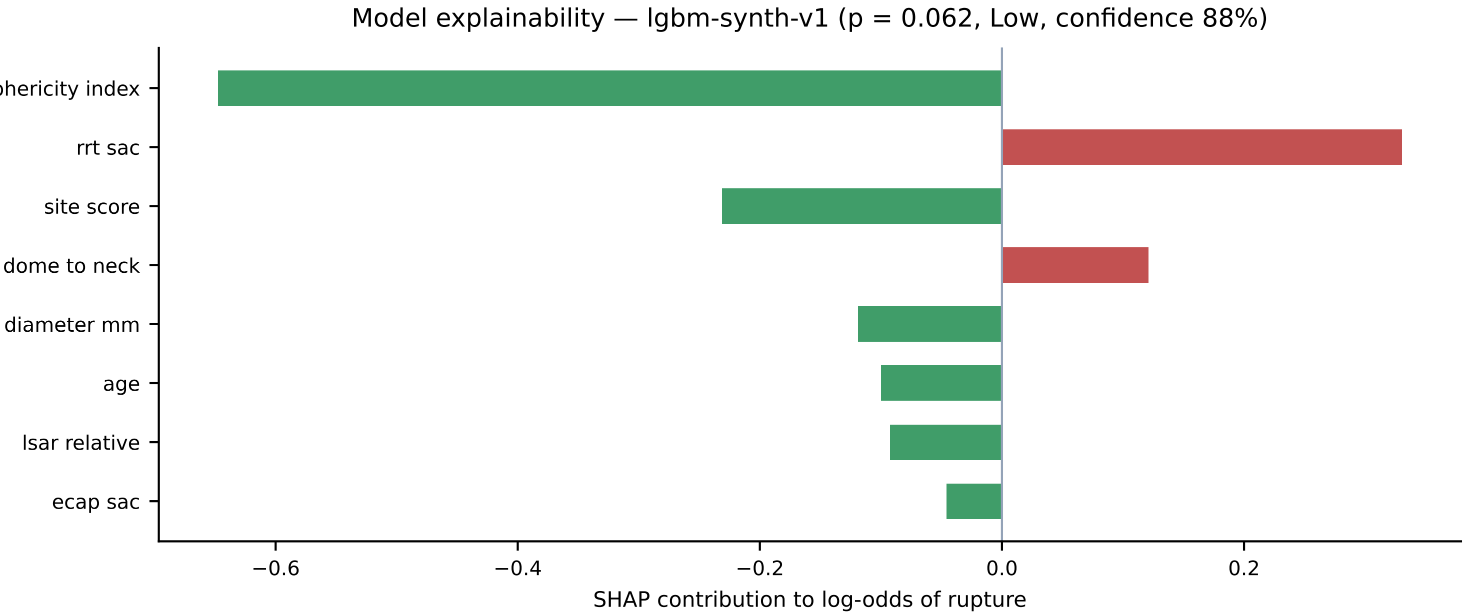
Zone Analysis and Model Explainability

Case PT-2026-0103





Assessment

Computational fluid dynamics analysis of case PT-2026-0103. Time-averaged wall shear stress at the aneurysm dome measured 0.346 Pa against 2.939 Pa in the parent artery (normalised WSS 0.118), with an oscillatory shear index of 0.008. 71.3% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 7.17 Pa^-1. Sac morphology measured from the reconstructed surface: maximum diameter 5.38 mm, neck 4.83 mm, aspect ratio 0.75, volume 63.4 mm^3. Composite Risk Index 39/100 (Low).

Model caveat: ILLUSTRATIVE ONLY. Not trained or validated on patient data. Demonstrates the feature/inference/explainability pipeline. Must not be used for clinical decision-making.